=> SML + LLM
=> DL (NN) + LLM
=> CV + LLM
=> LLM creating

} AI ning qaysi yo'nalishlari bo'limlaridan qatiy nazar (LLM) lardan foydalaniladi

1. API
2. chatbot
3. coding, reasoning

Layers => Attention, FFN → ~~serial~~ / Sequential

Normalization →
- MSE ×
- RMSNorm ✓ — masshtablaydi, markazlashtirmaydi — (2ta qadamni tashlaydi →)

Activation functions →
- ReLU ⊗
- SwiGLU
- GeLU ✓

Tokenization → tokens ↷ / → tokens ids (integer)
↓
Embeddings => integer to (vector)

Auto model => Pre-trained modellarini yuklab olib ishlatish imkonini beradi.

wte -> word token embeddings.
wpe -> word position embedding

Config.json -> model blueprint; loyihaviy chizmasi. Unda (no learned) knowledge lar ~~sog~~ saqlanadi. Config network shape, layerlar soni, attention va hokazolar saqlanadi

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Input -> Hidden layers -> Output
↓ tokenization embeddings
↓ Transformer layers, Norms, Architecture, Training...

gpt2

① Layer Norm -> MSE, MAE $eps = 10^{-5} = 0.00001$
elementwise_affine -> False o'rganilmaydi
True o'rganadi

=> Encoding da (SMAL) raqamlarga o'tib chiqqanlarchi hech narsa o'rganmaydi.

=> LLM da Norm, Embed, -> Encodingdan farqli o'rganadi. training ga o'xshash jarayon.

② Attention Layer -> raqamlar bir birini bilan aloqasi o'rganiladi

- Har bir LLM da o'zlari nom qo'yadi. (Chunki ularning ishlarini yashirish maqsadida. Qanchadir qismi (private) bo'ladi

input dimen, output_dimen berilədi.

dimensions -> hər bir tokeni ifadə olunacaq
rəqəmlərdən ibarət vectorun
uzunluğu

Anthropic API

Onlar bir birindən fərqlidi və
onun içəki texnologiyaları həm fərqlidi

user input.

① What is quantum computing?

↓

What | is | quantum | computing | ?

tokens.

① Request -> Tokenization
Embedding
Contextualization
Response <- Generation

Embeddings -> dimensions açıqlanmayıb.

what -> [0.12, 0.3, -0.5, ..., 0.25]
is -> [-0.3, 0.7, 0.33, ..., -0.45]
⋮
? -> [0.35, 0.255, -0.33, ..., -0.332].

quantum → possible meanings of "quantum"
Juda ham köplam maʼnolari mavjud.

→ ushbu sözning pozitioni muhim.

wte, wpe

$\underbrace{\text{Attention + FFN + LayerNorm + Residual Connections}}$

Contextualization

Output layer:
30% mechanics
23% information
18% scientist
10% data
10% supremacy
8% computer

Generate qilish kk

output tokens: Quantum mechanics is a form of computing. EOS

max_tokens → 100

Token 100 ga yetganda generate töxtaydi

EOS → javob qisqa bölsa özi töxtaydi.

End of sequence token